$f(x) = \ln(1+x^2) + x^3$; интервал: $[-1;1]$; $n=3$

$x_0 = -1$; $f(-1) = \ln(2) - 1$

$x_1 = 0$; $f(0) = 0$

$x_2 = 1$; $f(1) = \ln(2) + 1$

$\Rightarrow F(x_0, x_1) = \frac{0-(\ln 2 - 1)}{0-(-1)} = 1 - \ln 2$

$\Rightarrow F(x_1, x_2) = \frac{(\ln 2 + 1) - 0}{1-0} = \ln 2 + 1$

$F(x_0, x_1, x_2) = \frac{(\ln 2 + 1) - (1 - \ln 2)}{1-(-1)} = \ln 2$

$P_2(x) = (\ln 2 - 1) + (1 - \ln 2)(x+1) + \ln 2\,(x+1)x$

$P_2(x) = x + \ln 2\, x^2$

В узлах:

$x_0 = -1$; $P_2(-1) = \ln 2 - 1 = F(-1)$

$x_1 = 0$; $P_2(0) = 0 = F(0)$

$x_2 = 1$; $P_2(1) = 1 + \ln 2 = F(1)$

$\Rightarrow P_2 = F\big|_{x \in \{x_0, x_1, x_2\}}$

В серединах:

$x = -0{,}5$; $P_2(-0{,}5) = -0{,}5 + \ln 2 \cdot 0{,}25$

$F(-0{,}5) = \ln(1{,}25) + 0{,}125$

$\Rightarrow |P_2(-0{,}5) - F(-0{,}5)| \approx 0{,}675$

$x = 0{,}5$; $P_2(0{,}5) = 0{,}5 + \ln 2 \cdot 0{,}25$

$F(0{,}5) = \ln(1{,}25) + 0{,}125$

$\Rightarrow |P_2(0{,}5) - F(0{,}5)| \approx 0{,}325$